AQ5.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Consider a function $T: V \to W$, where V and W are the vector spaces and following statements:

Statement 1: $T(v_1 + v_2) = T(v_1) + T(v_2)$, where $v_1, v_2 \in V$.
Statement 2: $T(cv_1) = cT(v_1)$, where $x \in V$ and $c \in R$.

If T is a linear transformation then T satisfies
- ◯ statement 1 only.
- ◯ statement 2 only.
- ◯ both the statements 1 and 2.
- ◯ neither statement 1 nor statement 2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
both the statements 1 and 2.

1 point
Which of the following is/are a linear transformation.

- ☐ $T: R^2 \to R^3, T(x, y) = (x, x + y, y + 1)$
- ☐ $T: R^2 \to R^2, T(x, y) = (xy, x + y)$
- ☐ $T: R^2 \to R^3, T(x, y) = (x + y, y, 2x + y + 1)$
- ☐ $T: R^2 \to R^3, T(x, y) = (x + y, y, 2x + y)$
- ☐ $T: R^2 \to R^2, T(x, y) = (\frac{x}{2}, \frac{y}{2})$
- ☐ $T: R^2 \to R^2, T(x, y) = (\frac{x}{2}, \frac{xy}{2})$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T: R^2 \to R^3, T(x, y) = (x + y, y, 2x + y)$
$T: R^2 \to R^2, T(x, y) = (\frac{x}{2}, \frac{y}{2})$

1 point
Let $T: R^3 \to R^4$ be a linear transformation. Then which of the following options is /are true?

- ☐ $T(v_1 + v_2) = T(v_1) + T(v_2)$, where $v_1, v_2 \in R^3$.
- ☐ $T(0, 0, 0) = (0, 0, 0, 0)$
- ☐

$T(0,0,0) = (1,1,1,1)$

☐

$T(3(1,2,0)) = 3T(1,2,0)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$T(v_1 + v_2) = T(v_1) + T(v_2)$, where $v_1, v_2 \in R^3$.

$T(0,0,0) = (0,0,0,0)$

$T(3(1,2,0)) = 3T(1,2,0)$

1 point

Let $T: R^3 \to R^2$ be a linear transformation, defined as $T(x,y,z) = (x+y, y+z)$. Then which of the following options is /are true?

☐

$T(0,0,0) = (0,0)$

☐

$T(1,2,3) = (3,5)$

☐

$T(0,1,1) = (1,2)$

☐

$T(1,1,1) = (3,2)$

☐

T is injective.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$T(0,0,0) = (0,0)$

$T(1,2,3) = (3,5)$

$T(0,1,1) = (1,2)$

1 point

Choose the correct set of options:

☐

T is a one to one linear transformation if and only if there does not exist any $v \neq 0 \in V$ so that $T(v) = 0$.

☐

If $T: R^2 \to R^2$ is a surjective linear transformation, then it cannot be injective.

☐

If $T: R \to R$ is a linear transformation which is not injective, then T must be 0, i,e., $T(v) = 0$ for all $v \in R$.

☐

If there exists some non-zero vector $v \in V$, such that $T(v) = 0$, for a linear transformation $T: V \to W$, then T cannot be an isomorphism.

No, the answer is incorrect.
Score: 0
Accepted Answers:

T is a one to one linear transformation if and only if there does not exist any $v \neq 0 \in V$ so that $T(v) = 0$.

If $T: R \to R$ is a linear transformation which is not injective, then T must be 0, i,e., $T(v) = 0$ for all $v \in R$.

If there exists some non-zero vector $v \in V$, such that $T(v) = 0$, for a linear transformation $T: V \to W$, then T cannot be an isomorphism.

1 point

If $T: \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$, then which of the following options is true?

○ T is both one to one and onto.

○ T is one to one, but not onto.

○ T is onto, but not one to one.

○ T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is neither one to one nor onto.

1 point

If $T: \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x + y, x - y)$, then which of the following options is true?

○ T is both one to one and onto.

○ T is one to one, but not onto.

○ T is onto, but not one to one.

○ T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is both one to one and onto.

1 point

Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation, such that $T(2, 1) = (3, 2)$ and $T(1, 3) = (0, 0)$. Which of the following options is true?

○ $T(x, y) = \frac{1}{5}(3x, 2y)$

○ $T(x, y) = (x + 1, y + 1)$

○ $T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$

○ T cannot be determined from the given information.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$

1 point

Consider the linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ defined as $T(v) = Av$, where $v = \begin{bmatrix} x \\ y \end{bmatrix}$, and $A = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$. Which of the following options is true?

○ T is an isomorphism.

- ○ T is one to one, but not onto.
- ○ T is onto, but not one to one.
- ○ T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is an isomorphism.

Level 2

1 point

Let $S: V_1 \to V_2$ and $T: V_2 \to V_3$ be two linear transformations. Let us define $T \circ S: V_1 \to V_3$ by $T \circ S(v) = T(S(v))$. Choose the correct set of options.

- ☐ If $T \circ S$ is injective, then T must be injective.
- ☐ If $T \circ S$ is injective, then S must be injective.
- ☐ If $T \circ S$ is surjective, then T must be surjective.
- ☐ If $T \circ S$ is surjective, then S must be surjective.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $T \circ S$ is injective, then S must be injective.
If $T \circ S$ is surjective, then T must be surjective.

1 point

If $T: R^2 \to R^3$ is a linear transformation defined by $T(x, y) = (2x + 3y, 5x - y, x + 6y)$. Which of the following options is true?

- ○ T is both one to one and onto.
- ○ T is one to one, but not onto.
- ○ T is onto, but not one to one.
- ○ T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is one to one, but not onto.

1 point

If $T: R^3 \to R^3$ is a linear transformation defined as follows, $T: R^3 \to R^3$

$T(x, y, z) = (a_1 x + b_1 y + c_1 z, a_2 x + b_2 y + c_2 z, a_3 x + b_3 y + c_3 z)$. This can be represented as $T(v) = Av$, where

$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, and $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Which of the following options are true?

- [ ] T is injective if and only if the system of linear equations $Av = 0$ has a unique solution.
- [ ] T is injective if and only if $rank(A) = 3$.
- [ ] If the system of linear equations $Av = b$ for any $b \in R^3$, has a solution, then T must be surjective.
- [ ] If T is not surjective, then there exists some $b \in R^3$, such that the system of linear equations $Av = b$ has no solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is injective if and only if the system of linear equations $Av = 0$ has a unique solution.

T is injective if and only if $rank(A) = 3$.

If the system of linear equations $Av = b$ for any $b \in R^3$, has a solution, then T must be surjective.

If T is not surjective, then there exists some $b \in R^3$, such that the system of linear equations $Av = b$ has no solution.

Check Answers

Your score is: 0/12